Extended Benchmarks for PAQ-KV: MMDocRAG, Cross-Reader Baselines, and MMDocIR (Internal Companion Note)

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Abstract

This internal note extends the PAQ-KV evaluation along the three axes a reviewer of the main manuscript would ask for: a third question-answering dataset (MMDocRAG), the full baseline arm family on the three second readers (the main paper carries it only on Qwen3-VL-8B), and a direct below-page selection-quality measurement against MMDocIR's sub-page gold regions. Headline: on a fresh-documents-only MMDocRAG pin (62 documents, 585 queries; every document overlapping our development or sealed pins excluded), PAQ-KV@5% scores 0.4089 versus 0.3517 for the corrected-loader ColQwen2 at matched budget (+0.057, document-clustered $\mathrm{LB}_{95} = +0.040$), and is statistically indistinguishable from both full-context decoding and the gold-page oracle. The dispersed-evidence pattern of MMLongBench-Doc therefore reproduces on a third dataset; the conclusion's dispersed-vs-conversion-bound hypothesis survives its first out-of-benchmark test. The cross-reader panel adds that PAQ-KV stays significantly ahead of every training-free baseline (SnapKV-fresh/stale, CLIP, committed retrievers, random, truncation) on all four readers — including the readers where the trained retriever wins — and that SnapKV-fresh's own level falls across readers in exactly the order of the main paper's recovery gradient, independently confirming the attention-as-retriever boundary. On MMDocIR's sub-page gold, PAQ-KV's blocks cover 0.689 of gold-layout area at a realized 5.3% budget (random: 0.056), matching page-granularity selection's area at two-thirds of its realized cost while touching +0.140 more of the annotated layouts. Round 2 adds four things: a same-cohort re-run of the full baseline family on the development reader (replacing Table 1's daggered borrowings; the reader-independent arms reproduce the old panel to three decimals); a cap-sensitivity check on MMDocRAG (the contrast's direction and margin class are stable on the uncapped remainder, +0.035 [+0.019, +0.060]); and a *fourth* QA cell, MMDocIR-QA, where the result is a clean negative: ColQwen2-fresh 0.408 significantly outperforms PAQ-KV 0.365 (−0.043 [−0.065, −0.011]) on a retrieval-native question distribution. The four-cell picture — two dispersed-evidence wins, one conversion-bound tie, one concentrated-evidence loss — is a sharper test of the paper's dispersed-vs-concentrated hypothesis than three positives would have been, and it supports it. All results here are development-grade (no pre-frozen gates; single runs; the sealed sets were not touched).

1 Scope, Provenance, and the Overlap Audit

What this note adds. (i) §2: a complete arm panel on MMDocRAG (Dong et al. 2025b) with the frozen Qwen3-VL-8B stack; (ii) §5: the Table-4 baseline family (gold-page oracle, CLIP-fresh, ColQwen2-stale, random, truncation, SnapKV-fresh/stale) executed on Qwen3-VL-4B, Qwen2.5-VL-7B, and InternVL3.5-8B on both development pins, so the cross-reader table of the main paper gains the same comparator context as its Qwen3-VL-8B panel; (iii) §6: PAQ-KV's block selections scored directly against MMDocIR's (Dong et al. 2025a) expert-annotated sub-page layout regions, decoupling selection quality from QA conversion.

Evidence status. Everything in this note is development-grade: the harnesses reuse the study's frozen primitives (paqc_spike prefill/preview/select/decode, the corrected ColQwen2 loader, mmlb_scoring, and the document-cluster paired bootstrap paq.cluster_lb95, 10,000 draws, seed 0), but no bar was frozen before these runs, each cell was run once, and the pins were constructed on the same day as the runs. No sealed set was evaluated or read beyond document names (below).

Overlap audit and exclusion (disclosure). Both new benchmarks are corpus-adjacent to ours: MMDocIR was sourced from existing DocVQA collections including MMLongBench-Doc, and MMDocRAG was built over the MMDocIR document pool. A document-name audit against our pins found: MMDocRAG (223 documents) shares 75 documents with the development pins and **42 with the sealed pins**; MMDocIR (313 documents) shares 79 with the development pins and **45 with the sealed pins** — including, name-for-name, the entire sealed MMLongBench-Doc cell. Every document overlapping either the development or the sealed pins was excluded from the new pins before any experiment ran, so the sealed reserve remains untouched by development at the document level. The audit artifact is retained (research/newbench_docnames_audit_raw.json); only document names were read from the sealed pin file.

	MMDocRAG
PAQ-KV@5% (ours)	**0.4089**
ColQwen2-fresh (matched budget, corrected loader)	0.3517
SnapKV-fresh (page level)	0.1557
CLIP-fresh	0.1119
random	0.0368
truncation	0.0327
full-context (reference)	0.4018
gold-page oracle (reference; not strict)	0.4028
PAQ-KV − ColQwen2-fresh	$+0.057$ $[+0.040, +0.076]$
PAQ-KV − full-context	$+0.007$ $[-0.004, +0.018]$
PAQ-KV − gold-page oracle	$+0.006$ $[-0.009, +0.022]$
PAQ-KV − SnapKV-fresh	$+0.253$ $[+0.217, +0.288]$
Queries (decoded / paired)	585 / 585
Documents	62

Table 1: The MMDocRAG cell at the 62-document fresh-documents pin, nominal 5% budget, reader Qwen3-VL-8B (frozen stack). Contrast rows report Δ $[\mathrm{LB}_{95}, \mathrm{UB}_{95}]$ from the document-cluster paired bootstrap. All arms are complete (4,680/4,680 rows, zero skips). Development-grade: single run, no pre-frozen bar.

2 MMDocRAG: A Third QA Cell

2.1 Pin

From the 106 fresh MMDocRAG documents, 62 qualify under the study's standard pin protocol ($\geq$ 3 gold-bearing queries, $\leq$ 110 union pages; 30 dropped for length, 11 for query count, 3 lacked a PDF in the release). This is below the study's 80-document floor and is recorded as such (floors_met: false); all 62 are taken (seed-0 protocol). Queries are capped at 12 per document by seed-0 sampling (28 documents hit the cap; 585 of 1,160 available queries kept) to keep the cell comparable in size to the existing pins — a deviation from the main pins, which never needed a cap, and it is disclosed here. Pages are rendered from the released PDFs at 144 DPI (matching the existing pins' pixel scale); gold pages are the union of page_id over each query's gold_quotes (the 0-based convention was verified by text-fragment matching on 32/32 probes and visually). Answers are scored by the study's shared document-VQA evaluator against answer_short. Mean gold pages per query is 1.68; mean pages per document 27.9 (median 12, max 106).

2.2 Results

Table 1 reports the full panel. Three observations.

The improvement over the trained retriever reproduces. PAQ-KV@5% scores 0.4089 vs. 0.3517 for corrected-loader ColQwen2 at matched budget: $+0.057$ with document-clustered $\mathrm{LB}_{95} = +0.040$. The interval excludes zero with the same margin class as the MMLongBench-Doc development result ($+0.077$, $\mathrm{LB}_{95} = +0.040$) and the sealed confirmation ($+0.076$, $\mathrm{LB}_{95} = +0.029$). MMDocRAG was designed for retrieval-augmented document QA with cross-modal, multi-page evidence chains (52% cross-page), and the result is consistent with the main paper's evidence-dissociation reading: this is a dispersed-evidence cell, and PAQ-KV behaves on it exactly as the hypothesis predicts.

Indistinguishable from full context and from the oracle. PAQ-KV sits $+0.007$ $[-0.004, +0.018]$ against its own full-context read and $+0.006$ $[-0.009, +0.022]$ against the gold-page oracle. As in the main paper, the oracle interval permits material differences in both directions at this sample size; we claim parity within noise, not equality. Notably the oracle here is barely above full context (0.4028 vs. 0.4018), consistent with MMDocRAG's evidence being genuinely dispersed; handing the reader only the annotated gold pages does not help it.

The eviction-style and off-the-shelf baselines are far behind. Page-level SnapKV-fresh reaches 0.1557 (PAQ-KV margin $+0.253$, $\mathrm{LB}_{95} = +0.217$), CLIP-fresh 0.1119, and the random/truncation controls 0.037/0.033. This mirrors the main paper's Appendix panel ordering and supports treating the trained retriever as the binding comparator on this cell as well.

Round-2 additions: two more baselines and the cap-sensitivity check. ColQwen2-stale (commit-once aggregate of the corrected-loader scores) collapses here as everywhere (0.1749; PAQ-KV $+0.234$ $[+0.204, +0.263]$), and a BM25 page retriever over the PDFs' born-digital text layer reaches 0.2793 (PAQ-KV $+0.130$ $[+0.106, +0.154]$). The cap-12 query subsample was also stress-tested: decoding the 575 uncapped remainder queries on the same 62 documents gives PAQ-KV − ColQwen2 $= +0.035$ $[+0.019, +0.060]$ (pooled over all 1,160 queries: $+0.046$ $[+0.032, +0.064]$) — the same direction and margin class as the capped cell, so the cap did not manufacture the result, though the capped point estimate ($+0.057$) is somewhat larger than the remainder's.

Caveats. 62 documents is below the study's power floor; the cap-12 query subsample is a protocol deviation (now bounded by the sensitivity check above); the cell was evaluated once with no pre-frozen bar; and MMDocRAG's expert annotations were built over MMDocIR's document pool, so its domain mix (heavier on financial reports and academic papers) differs from MMLongBench-Doc's. A sealed-style confirmation was attempted and found **structurally impossible**: the 41 never-decoded fresh documents are unused precisely because none qualifies under the frozen pin protocol (30 exceed 110 pages, 11 have fewer than 3 gold-bearing queries); the options are laid out in research/design_notes/paq_mmdocrag_reserve_5a_2026-07-26.md and require a human decision.

3 MMDocIR-QA: A Fourth QA Cell, and a Negative

MMDocIR's questions carry answers and gold pages, so the same 80-document fresh-only pin used for the selection-quality analysis (§6) supports a full QA cell (538 queries; answers verified scoreable by the study's evaluator on a 30-probe gate before any GPU was spent; one query has an empty gold answer and scores 0 for every arm, disclosed). Table 2 carries the numbers in its MMDocIR-QA column.

The trained retriever wins this cell. ColQwen2-fresh scores 0.4080 vs. PAQ-KV 0.3654: -0.043 $[-0.065, -0.011]$, the first cell in the study where the retriever significantly outperforms PAQ-KV on the development reader. PAQ-KV remains statistically tied with full context (-0.013 $[-0.049, +0.012]$) and far above every training-free baseline (SnapKV-fresh $+0.167$, CLIP $+0.225$, random $+0.285$, all intervals excluding zero), so the mechanism works here too — the retriever is simply better at this cell's questions.

Why, and why this helps the paper. The cell's anatomy points one way: the gold-page oracle (0.4608) towers over full context (0.3785), and ColQwen2 itself beats full context — large selection headroom, of the kind a strong retriever captures. MMDocIR's questions were written *for retrieval evaluation*, each targeting specific annotated layouts: a concentrated-evidence, retrieval-native distribution. The four QA cells now split exactly along the hypothesis the paper's conclusion offers: dispersed-evidence cells are PAQ-KV wins (MMLongBench-Doc $+0.077$; MMDocRAG $+0.057$), the conversion-bound cell is a tie (LongDocURL $+0.019$, n.s.), and the concentrated-evidence cell is a retriever win (MMDocIR-QA -0.043). We consider this negative a scope-defining result worth reporting next to the wins; note also the round-1 selection-quality finding that PAQ-KV's *selector* covers this cell's gold layouts well (§6) — the loss is in what a 5% block view converts, not in finding the evidence.

Consolidated main table. Table 2 assembles all five columns (LDU, MMLB development, MMLB sealed, MMDocRAG, MMDocIR-QA) from verdict artifacts. The sealed column carries only the two arms that were sealed-evaluated, by design.

4 Same-Cohort Baselines on the Development Reader

Table 1 of the manuscript previously borrowed five baseline rows from the page-level 903-query cohort (pre-cache-fix, daggered as context). Round 2 re-ran the full arm family on the development reader on both 80-document pins, same cohort and read path as the main below-page rows. Two facts matter. First, the reader-independent arms (CLIP, ColQwen2-stale, random, truncation, oracle) reproduce the old panel to three decimals (e.g. LDU oracle 0.6092 vs. 0.609; ColQwen2-stale 0.2160 vs. 0.216), which cross-validates both cohorts and retires the daggers without changing any conclusion. Second, SnapKV-fresh — the one arm whose score depends on the preview resolution — roughly doubles when scored from the full-resolution prefill ($0.3243/0.4892$ MMLB/LDU vs. $0.157/0.298$ coarse), landing just below the trained retriever and significantly below PAQ-KV on MMLB (-0.080 $[-0.119, -0.043]$) and marginally so on LDU (-0.042 $[-0.092, +0.001]$). The manuscript's "strongest training-free baseline" framing should therefore cite these same-cohort SnapKV values, not the coarse-preview ones.

5 Cross-Reader Baseline Panel

The main paper's cross-reader table compares PAQ-KV only against ColQwen2-fresh and full context on the three second readers. This section adds the remaining arm family at the same $b = 5\%$ budget, with page-set semantics identical to the Qwen3-VL-8B panel (same cached CLIP/ColQwen2 page scores, same commit-once aggregation over the first $M = \min(3, k)$ queries, same seed-0 random and truncation controls, gold-page oracle read directly) and the read path identical to each port's ColQwen2 comparator arm, so every number is within-reader comparable. SnapKV arms are computed from each reader's *own* question-preview attention (the port's preview machinery), aggregated to pages with the panel's pooling, so SnapKV-fresh/stale are genuinely per-reader rather than proxied from the development reader.

Table 3 reports the panel (all six reader×cell runs complete: 907 queries × 7 arms per reader, zero skipped rows). Four observations.

PAQ-KV stays ahead of every training-free baseline on every reader, including where it loses to the trained retriever. SnapKV-fresh is the strongest non-retriever arm in all six cells, and it is significantly below PAQ-KV in every one: -0.061 $[-0.096, -0.025]$ / -0.083 $[-0.144, -0.031]$ on Qwen3-VL-4B (MMLB/LDU), -0.143 on both Qwen2.5-VL cells, and -0.099 / -0.142 on InternVL3.5-8B. So even on the readers where PAQ-KV does not match ColQwen2, the ordering *within* the training-free family is preserved: the full method (below-page blocks, masked decode over the retained cache) never loses to its page-level SnapKV-style reduction.

The SnapKV-fresh gradient independently confirms the architecture boundary. SnapKV-fresh here is precisely "the reader's own question-preview attention selecting pages," so its absolute level is a direct read-out of each reader's attention-as-retriever quality. It falls from 0.303 (Qwen3-VL-4B) to 0.127 (Qwen2.5-VL-7B) to 0.089 (InternVL3.5-8B) on MMLongBench-Doc, and $0.413 \rightarrow 0.252 \rightarrow 0.077$ on LongDocURL — the same ordering as the main paper's recovery gradient ($0.87/0.70/0.53$), measured with an independent arm at page granularity. The cross-reader reversal of PAQ-KV on InternVL is therefore not an artifact of the block machinery: the selection signal itself degrades in this order.

Reversibility generalizes across readers and scorer families. Committing either scorer once collapses it on every reader: SnapKV-stale lands at 0.034 to 0.133 (versus 0.077 to 0.413 fresh) and ColQwen2-stale at 0.107 to 0.211 (versus 0.262 to 0.473 fresh). The main paper established fresh-over-stale on Qwen3-VL-8B; this panel shows the multi-query staleness penalty is a property of the regime on all four readers tested, not of the development reader.

Oracle headroom grows as readers weaken, and InternVL's failure is selection, not conversion. Unlike Qwen3-VL-8B (statistically indistinguishable from the oracle on MMLB), every second reader sits significantly below its own gold-page oracle ($+0.052$ to $+0.143$ oracle-over-PAQ-KV across the six cells). On

Arm	LDU	MMLB	MMLB-sealed	MMDocRAG	MMDocIR-QA
PAQ-KV@5% (ours)	0.5315	0.4042	0.3957	0.4089	0.3654
ColQwen2-fresh (matched budget)	0.5124	0.3282	0.3202	0.3517	0.4080
SnapKV-fresh	0.4892	0.3243	–	0.1557	0.1987
SnapKV-stale	0.1873	0.0902	–	–	–
CLIP-fresh	0.1549	0.1624	–	0.1119	0.1400
ColQwen2-stale	0.2160	0.1365	–	0.1749	0.1583
BM25	0.4030	0.2109	–	0.2793	–
CAPS-style expert head (page level)	0.5182	0.3567	–	–	–
random	0.0675	0.0218	–	0.0368	0.0807
truncation	0.0329	0.0410	–	0.0327	0.0876
full-context (reference)	0.5456	0.3965	–	0.4018	0.3785
gold-page oracle (ref.; not strict)	0.6092	0.4300	–	0.4028	0.4608

Table 2: Consolidated arm panel across all cells (accuracy at nominal b=5%, reader Qwen3-VL-8B). LDU/MMLB baselines are the round-2 *same-cohort* re-runs on the 80-document development pins (post-cache-fix, read path identical to the cached comparator rows); the reader-independent arms reproduce the earlier page-level panel to three decimals, while SnapKV-fresh is now scored from the full-resolution prefill attention (the same prefill PAQ-KV uses) rather than the earlier coarse-render preview, which raises it substantially and is the fair same-cohort form. The sealed column reports the frozen sealed verdict's two arms only; its other cells are empty by design. "–" cells are structurally absent (arm not defined or not run for that cell). Development-grade except the sealed column.

InternVL3.5-8B the oracle is also significantly above *full context* ($+0.052$ $[+0.013, +0.094]$ MMLB; $+0.045$ $[+0.007, +0.088]$ LDU): handing that reader exactly the gold pages beats letting it read everything, so evidence conversion is not its bottleneck — finding the evidence is. This sharpens the main paper's scope condition: where the host's attention retrieves poorly, a stronger selector (or the trained retriever) recovers real accuracy that PAQ-KV leaves on the table.

6 MMDocIR: Below-Page Selection Quality

MMDocIR provides expert gold at *layout* granularity (paragraphs, tables, figures, charts as boxes on the page), which is ground truth at exactly PAQ-KV's below-page granularity — something neither of the paper's QA benchmarks offers. On a fresh-documents-only pin (80 documents, 538 queries, 641 gold layouts; same exclusion audit as above), we score PAQ-KV's selected 64-token blocks geometrically against the gold boxes: block token spans are mapped through the reader's own patch grid to page-pixel rectangles, and we measure gold-layout area coverage, layout hit rate, and gold-page recall at $b \in \{5, 10\}$%, against random-block, truncation-block, and page-granularity comparators at matched token budget. No decoding is involved, so this isolates the selector from QA conversion entirely.

Table 4 reports the analysis (538 queries $\times$ 4 arms $\times$ 2 budgets plus the reference, 4,842 rows, zero skips; reader Qwen3-VL-8B, frozen stack). Three observations.

The selector finds sub-page evidence. At a realized 5.3% of context, PAQ-KV's blocks cover 0.689 of gold-layout *area*, touch 0.834 of gold layouts, and reach 0.903 gold-page recall. Random blocks at the same budget cover 0.056 (paired contrast $+0.632$ $[+0.595, +0.683]$); truncation covers 0.079. This is the first conversion-free measurement of the study's selection signal against human-annotated sub-page ground truth, and it is unambiguous.

Below-page granularity buys reach, not raw area. The page-granularity arm (identical attention scores aggregated to pages) covers the same gold area (0.693 vs. 0.689; -0.004 $[-0.070, +0.044]$, n.s.) but needs a realized 7.7% of context to do it (whole-page rounding), and at that higher spend still touches *fewer* distinct gold layouts ($+0.140$ $[+0.097, +0.170]$ in PAQ-KV's favor) and fewer gold pages ($+0.209$ $[+0.126, +0.277]$). Blocks spread the same budget across more of the question's evidence — exactly the behavior the main paper's cross-page QA advantage suggested, now measured geometrically.

The whole-gold-pages reference prices the annotation. Selecting every gold page in full costs a realized 5.6% of context on this pin — essentially PAQ-KV's own spend — so at matched budget a perfect page-level selector could reach coverage 1.0. PAQ-KV recovers 69% of that area (and 83% of the layouts) without any supervision, from the reader's own attention. The remaining gap is the selection headroom that the QA oracle contrasts of §5 price in accuracy terms.

7 What This Changes for the Main Manuscript

Two further round-2 facts belong here. The ColQwen2 training-distribution audit (research/design_notes/colqwen_training_overlap_audit_2026-07-26.md) found no confirmed document-level overlap between the comparator's training mixture and any of the four evaluation corpora, and no provable disjointness for the web-sourced portions either; the honest manuscript sentence is that the sealed win beats a comparator that is *task-trained and genre-matched* (same source families: financial reports, arXiv-figure data, government and industry web PDFs) while the reader has no retrieval training. And the sealed-style MMDocRAG confirmation proposed in round 1 is structurally impossible under the frozen protocol (no fresh document qualifies; §2), which makes the MMDocIR-QA cell's independent test — negative as it is — the de facto corroboration lane for the four-cell hypothesis.

The additions map onto the manuscript as follows. (i) The MMDocRAG cell is the natural third column for Table 1 (or a separate results table if the 62-document pin and cap-12 protocol deviations argue against mixing cohorts): it directly tests the conclusion's offered hypothesis and supports it, with the same margin class as the sealed MMLongBench-Doc result. (ii) The cross-reader baseline panel upgrades §5.4's three-arm comparison to the main results' panel discipline, preempts the "did you only compare against weak baselines on the other readers?" review question, and contributes a genuinely new mechanistic datum (the SnapKV-fresh gradient) for the architecture-boundary claim. (iii) The MMDocIR analysis gives the below-page granularity claim its first direct, conversion-free measurement, and the "reach, not raw area" framing cleanly explains why block selection wins QA despite losing coverage-weighted recall (the main paper's dissociation). We recommend keeping all three as appendix or companion material rather than headline claims until a sealed-style confirmation exists for MMDocRAG; its remaining fresh documents (44 unused, plus the uncapped queries) could seed such a reserve if promoted.

Arm	LDU	MMLB
Qwen3-VL-4B (same architecture)		
PAQ-KV@5% (ours; cached)	0.4966	0.3640
ColQwen2-fresh (cached)	0.4908	0.3336
full-context (ref.; cached)	0.5024	0.3480
SnapKV-fresh	0.4133	0.3034
SnapKV-stale	0.1332	0.0873
CLIP-fresh	0.1545	0.1805
ColQwen2-stale	0.2105	0.1576
random	0.0684	0.0296
truncation	0.0322	0.0423
gold-page oracle (ref.)	0.5839	0.4373
Qwen2.5-VL-7B (same family, prior gen.)		
PAQ-KV@5% (ours; cached)	0.3953	0.2698
ColQwen2-fresh (cached)	0.4728	0.2623
full-context (ref.; cached)	0.4558	0.3237
SnapKV-fresh	0.2519	0.1265
SnapKV-stale	0.0574	0.0342
CLIP-fresh	0.1591	0.1354
ColQwen2-stale	0.2047	0.1245
random	0.0609	0.0253
truncation	0.0343	0.0394
gold-page oracle (ref.)	0.5380	0.3219
InternVL3.5-8B (cross-family, Qwen3 LM)		
PAQ-KV@5% (ours; cached)	0.2186	0.1876
ColQwen2-fresh (cached)	0.3081	0.2542
full-context (ref.; cached)	0.2961	0.2637
SnapKV-fresh	0.0770	0.0889
SnapKV-stale	0.0537	0.0492
CLIP-fresh	0.1121	0.1498
ColQwen2-stale	0.1549	0.1068
random	0.0675	0.0377
truncation	0.0293	0.0385
gold-page oracle (ref.)	0.3408	0.3159

Table 3: Baseline arm family on the three second readers at the 80-document development pins (LDU = LongDocURL, MMLB = MMLongBench-Doc), $b = 5\%$. The PAQ-KV/ColQwen2-fresh/full-context rows are the cached values from the cross-reader ports (identical to the main paper's Table); the remaining arms are new runs with matched page-set semantics. Development-grade.

Arm	layout cov.	layout hit	page recall	realized b
nominal budget $b = 0.05$				
PAQ-KV blocks (ours)	**0.689**	0.834	0.903	5.3%
page level (same scorer)	0.693	0.694	0.694	7.7%
random blocks	0.056	0.141	0.449	5.3%
truncation blocks	0.079	0.087	0.095	5.4%
nominal budget $b = 0.1$				
PAQ-KV blocks (ours)	**0.778**	0.895	0.960	10.3%
page level (same scorer)	0.791	0.792	0.792	12.2%
random blocks	0.114	0.273	0.720	10.2%
truncation blocks	0.125	0.127	0.138	10.3%
whole gold pages (upper ref.)	1.000	1.000	1.000	5.6%

Table 4: Selection quality against MMDocIR's sub-page gold layouts on the 80-document fresh pin (538 queries). "Coverage" is the fraction of gold-layout area covered by selected blocks on gold pages; "hit" the fraction of gold layouts touched at all; "gold-page recall" the fraction of gold pages receiving any block. Development-grade.

References

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